Lesson 16 Smallest Counterexample

In Lessons 14 and 15, we learned the technique of induction. In this lesson, we introduce **proof by smallest counterexample**, which is essentially a combination of induction and contradiction. We redo Proposition 1 of Lesson 14.

Proposition 1 For every natural number n,

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2} \quad (*)$$

Proof Suppose that the statement is false. Then there is a smallest counterexample, i.e. a smallest positive integer $n = k$ such that the statement $(*)$ is not true. Note that $k \neq 1$ because

$$1 = \frac{1(1+1)}{2}$$

So $k > 1$, and $k - 1 > 0$. Then the statement is true for $k - 1$, since $k - 1$ is smaller than the smallest counterexample and cannot be a counterexample. Hence,

$$1 + 2 + \cdots + (k-1) = \frac{(k-1)k}{2}$$

Then, for $n = k$:

$$1 + 2 + \cdots + (k-1) + k = \frac{(k-1)k}{2} + k = \frac{k}{2}\big((k-1) + 2\big) = \frac{k}{2}(k+1)$$

This contradicts that k is a counterexample. So the statement $(*)$ is true.

If we compare the above proof to the proof in Lesson 14, we see that the steps in a proof by smallest counterexample closely resemble the steps in an induction proof.

- In induction, we show that if a statement is true for $n = k$, then it is true for $n = k + 1$.
- In proof by smallest counterexample, we suppose that the statement is false, and consider the smallest counterexample $n = k$. Then the statement is true for $n = k - 1$. We show that the statement is true for $n = k$, which contradicts the assumption that $n = k$ is a counterexample.
- In induction, we go from $n = k$ to $n = k + 1$, and in proof by counterexample, we work from $n = k - 1$ to $n = k$.
- In induction, we show that the statement is true for the base case $n = a$ for a statement which claims to be true for $n \geq a$. In proof by smallest counterexample, we show that the lowest counterexample $n = k$ is NOT the lowest case but showing that the statement is true for the $n = a$. We must use a, not a higher integer.
- The proof by induction and the proof by smallest counterexample are based on the well-ordering principle of natural numbers:

  Every set of integers bounded below contains a least element.

In Lesson 15, we used strong induction to prove that every integer greater than 2 has a prime factorization. This is only half the Fundamental Theorem of Arithmetic; the other half is the uniqueness of prime factorization. We will prove the uniqueness of prime factorization using smallest counterexample.

Fundamental Theorem of Arithmetic Every integer $n > 1$ has a unique prime factorization, that is, n can be written as a product of primes in exactly one way: $n = p_1 p_2 \dots p_k$, where $p_1, p_2, \dots, p_k$ are primes (which may occur in any order).

Proof Suppose that not every n has a unique prime factorization.
Let x be the smallest integer greater than 1 which has two different factorizations:

$$x = p_1 p_2 \cdots p_j, \qquad x = q_1 q_2 \cdots q_l$$

where p_i's and q_i's are primes.
We know that x is not a prime because then it has only one prime factor, itself.
So, $j > 1$ and $l > 1$.
Next, $p_1 | x$ implies that $p_1 | q_1 q_2 \dots q_l$.
Since p_1 is prime, $p_1 | q_i$ for some i, and since q_i is prime, $p_1 = q_i$.
If we divide x by p_1 in one product and divide x by q_i in another product, then

$$p_2 p_3 \cdots p_j = q_1 \cdots q_{i-1} q_{i+1} \cdots q_l$$

(we still have two products because $j > 1$ and $l > 1$).
Now $m = p_2 p_3 \cdots p_j < x$, but m has two prime factorizations, contradicting that x is the smallest integer greater than 1 which has two different factorizations!